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Method for dynamic video compression in an ad delivery ecosystem

Abstract

Example embodiments of the disclosure may provide a recompressor system that configures dynamic Video Ad Serving Template (VAST) tags enabling seamless delivery of mezzanine-level quality video ads of the highest resolution compatible with the user's system requirements. Aspects of the disclosure relate to reprocessing of received VAST tag, parsing the tag to identify the location of the video of highest resolution, accessing the video of highest resolution and forwarding it to a compression engine (Euclid's in this case). Compression engine re-compresses video in one or more formats and sends back location(s) of the videos. Receiving back one or more video locations and an updated VAST tag and forwarding that.

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Background/Summary

RELATED APPLICATION(S) (1) This application claims the benefit of U.S. Provisional Application No. 63/479,944, filed on Jan. 13, 2023, the entire teachings of which are incorporated herein by reference. (2) This application incorporates by reference the entire teachings and disclosures of U.S. application Ser. No. 17/144,924, filed Jan. 8, 2021, now U.S. Pat. No. 11,350,105, issued May 31, 2022, which is a continuation of U.S. application Ser. No. 16/926,089, filed Jul. 10, 2020, now U.S. Pat. No. 11,159,801 issued Oct. 26, 2021, which is a continuation of U.S. application Ser. No. 16/420,796, filed May 23, 2019, now U.S. Pat. No. 10,757,419, issued Aug. 25, 2020, which is a continuation-in-part of International Application No. PCT/US2017/067413, which designated the United States and was filed on Dec. 19, 2017, and which claims the benefit of U.S. Provisional Application No. 62/452,265 filed on Jan. 30, 2017. (3) The entire teachings of the above referenced applications are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to video compression of advertisements and the preparation of extensible markup language (XML), markup codes, scripting codes and control programing languages within an advertising ecosystem. These methods are also applicable to general video streaming content on the internet.

BACKGROUND

(2) Video compression can be considered the process of representing digital video data in a form that uses fewer bits when stored or transmitted. Video encoding can achieve compression by exploiting redundancies in the video data, whether spatial, temporal, or color-space. Video compression processes typically segment the video data into portions, such as groups of frames and groups of pels, to identify areas of redundancy within the video that can be represented with fewer bits than required by the original video data. When these redundancies in the data are exploited, greater compression can be achieved. An encoder can be used to transform the video data into an encoded format, while a decoder can be used to transform encoded video back into a form comparable to the original video data. The implementation of the encoder/decoder is referred to as a codec.

(3) Online advertising buyers and sellers use multiple protocols to describe and communicate, transact, manage, compress and deliver online video data. Online buyers, for example, may compress their video advertisements in various formats before adding them to an advertising network so that they are playable on a variety of devices, including televisions, mobile phones, tablets, watches, and computers. Buyers often take a generic approach to setting the video settings, as a result, the advertisement is not optimized for the device receiving and playing the advertisement. This is because consumer devices differ in screen size, computational capabilities and supported protocols.

SUMMARY

(4) Embodiments of the present disclosure can provide mezzanine compression that reduces the video bitrate (bandwidth) for encoded video ads. Aspects of the disclosure can enable encoded video-based ads to be customized for performance. In some embodiments, the video-based ads can be customized for specific devices and/or for specific bandwidth conditions. In some embodiments, the video-based ads can be customized for optimal performance.

(5) A content delivery network (CDN) may be a cluster of geographically distributed servers that speed up the delivery of web content by bringing it closer to where users are. Data centers across the globe use caching, a process that temporarily stores copies of files, so that a client device can access internet content from a web-enabled device or browser more quickly through a server that is in a relatively close geographic location as the client device. CDNs cache content including web pages, images, videos, and ad in proxy servers near to the client device's physical location to improve content delivery speed at the client device.

(6) Advertisers use a protocol to manage, describe, communicate, bid, and transmit video advertisements. Two popular protocols for this among buyers, ad networks, sellers, bidders, and delivery networks are VAST (Video Ad Serving Template) and VPAID (Video Player-Ad Interface Definition). This present disclosure describes the methods for dynamically adjusting video settings

and for dynamically updating the VAST and VPAID for each impression.
(7) Example embodiments herein provide techniques for providing video advertisements in different formats and sizes before and after they have been introduced into the digital ad ecosystem. A method for video compression through VAST and VPAID processing, based on marked up languages or scripting languages and digital video streams, to change the stream size, viewing quality, and format of a video. A VAST URL is passed to the new method, and the ad video streams are reprocessed into a new format, and the existing VAST and VPAID data elements are preserved and presented to the device so that all the tracking information is preserved.
(8) One exemplary aspect for the method disclosed is that this method can be applied at the ad server for buyers, or the demand side platform, or the supply side platform, or the ad server for sellers, or the client display device.
(9) In another exemplary aspect for the method disclosed is that the method can transform video ads to a wide variety of formats for a wide variety of devices. Device specific computation and supported video formats features are used to optimize the video format and settings for the targeted consumer device. Optimizing the video format, stream size and frame size maximizes the user experience.
(10) In another exemplary aspect for the method disclosed is that the optimized video format for the device can reduce the probability of network jitter causing buffering and video anomalies.
(11) In another exemplary aspect of the method disclosed is that Brands, Agencies, Ad Networks, Ad Servers, and Demand Side Platforms can stop sending generic device formatted videos to the digital ad ecosystem where this method is implemented. These entities can instead distribute mezzanine-level, high bit rate ad streams to the digital ad ecosystem and let this method prepare ad streams that better match the limits of a network's bandwidth, computational capability of a device and the device's screen size.
(12) Aspects of the disclosure relate to a dynamic reencoding system that intercepts VAST tags and replaces a video pointer address in the VAST tag with an address to a transformed video that improves transmission and/or video quality with respect to the client. A client device may be configured to communicate via a session handler with at least one server in a content delivery network. The at least one server may be configured to deploy a first video encoding for delivery at the client device via a first VAST tag transmitted via the session handler to the client device. A recompressor system may be configured to process the first VAST tag to identify the first video encoding. The recompressor system may be configured to recompress the first video encoding using bitrate encoding quality metrics to dynamically optimize the first video encoding for the client device such that the first video encoding is transformed into a first mezzanine encoding. The recompressor system may be configured to dynamically reconfigure the first VAST tag resulting in a transformed vast tag identifying the first mezzanine encoding while preserving information about the client device from the VAST tag including existing VAST and VPAID data elements so that all tracking information is preserved. The at least one server may be configured to respond to a request to deploy the first video encoding to the client by verifying that the first video encoding is in the video cache of the CDN and responding by forwarding the transformed VAST tag identifying the first mezzanine encoding to the client device via the session handler. The client device may be configured to respond to the transformed VAST tag by processing at least a portion of the first mezzanine encoding via the transformed VAST tag.
(13) In an embodiment, the disclosure provides a dynamic VAST tag encoding system. The system can dynamically rebuild VAST tags to ensure that videos being deployed to a user device are at optimal quality.
(14) In an example implementation, the system may process media loading media on a user device that performs a HTTP GET for an ad from an ad server. A demand side server, optionally, joins an ad auction and bids and wins the auction. The demand side server or a supply side ad server or any server may be configured to transmit video to client devices via calling a recompression server API.

That server may be configured to pass a VAST URL to a recompression server, which adds or updates a record in a cache with a key value pair. In response, the recompression server may build and stores a new VAST XML and responds with a new VAST URL to the new VAST XML.
(15) The demand side server or supply side ad server or any server may transmit video fills in URL macro fields. The specified server be configured to respond to the device with a VAST URL. The device may be configured to fill in macros on VAST URL. The device may be configured to fill in the HTTP headers with hardware and software information during a HTTP GET process to send a request to the demand server. The device may perform a HTTP get on the VAST URL with macros. The VAST rewriter server may perform a check of the cache for a compressed ad with a key value pair. The server can be configured to copy macros to original VAST URL so that the demand server may be called. The server responds with original XML and optionally a new ad video. In response, the VAST rewriting server may be configured to process a VAST XML to find the mezzanine URL for recompression.
(16) The VAST rewriting server may be configured to read the cache to get the replacement videos with the mezzanine URL. The VAST rewriting server, in this example scenario, may be configured to determine if there is a cache miss (e.g., determines the cache is empty and therefor a cache miss). The VAST rewriting server responds to the cache miss at the device by passing the original VAST XML and use original VAST XML, not the new VAST XML in this ad request.
(17) The VAST rewriting server performs a message queue to compress the video asynchronously. The recompression server HTTP gets the ad stream to be recompressed to mezzanine quality. The demand content delivery network (CDN) responds with the ad stream and recompression server starts compression of the ad. The recompression server posts the newly compressed ad stream(s) to the CDN. In a further embodiment, the recompression server updates cache with new encoded video data with a key value pair; and the client device performs a HTTP Get ad stream with original ad video, because this is a cache miss scenario; and the device streams the original uncompressed ad; and consumer sees the original ad.
(18) In the dynamic VAST tag encoding system, the VAST XML regenerating process may include original VAST XML data, new VAST XML data, and original video while the new video is recompressed, reformatted, repackaged or transformed from on codec to another to be used in future ad requests.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are included to provide further understanding and are incorporated in and constitute a part of this specification, illustrate disclosed embodiments and together with the description explain the principles of the disclosed embodiments in the drawings enclosed.
(2) The foregoing will be apparent from the following more particular description of example embodiments, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating embodiments.
(3) FIG. **1** is an illustration of an example of a high-level architecture for an online video advertisement system according to an embodiment.
(4) FIG. **2** is an illustration of an example as to how digital advertising works without the disclosed methods according to an embodiment.
(5) FIG. **3** is an illustration of an example as to how disclosed methods work in a cache miss scenario.
(6) FIG. **4** is an illustration of an example as to how disclosed methods work in a cache hit scenario

according to an embodiment.
(7) FIG. **5** is a schematic diagram of a computer network environment in which embodiments are deployed according to an embodiment.
(8) FIG. **6** is a block diagram of the computer nodes in the network of FIG. **5** according to an embodiment.
DETAILED DESCRIPTION
(9) A description of example embodiments follows.
(10) Typically, buyers of encoded video ads compress their own ads via an ad host server. For example, this can be done using tools, such as Google Analytics 360 digital video or Google Apps Manager (GAM). Google Analytics 360 includes a variety of tools such as: Analytics, Tag Manager, Optimize, Data Studio, Surveys, Attribution, and Audience Center. Video ad compression systems often compress the video data after the content has been introduced to the ad ecosystem and tagged.
(11) Certain example embodiments of the present disclosure include a recompressor system that can enable customized video compression for CSAI (Client-Side Ad Insertion). The recompressor system may provide dynamic device and bandwidth compression optimization based on user device-specific requirements and bandwidth-specific constraints. The recompressor system, for example, may be calibrated to increase ad performance and reduce stream size, buffering and errors, while improving fill rates, load time completion rates, and click-throughs. The recompressor system can enable video compression optimization tailored to devices, codec formats, and player screen dimensions, ultimately resulting in ad deliver of mezzanine-level quality ads. With example embodiments of the recompressor system, publishers can deliver ads from their own content delivery network (CDN), enabling the recompressor system to provide the publishers with optimization to control the ad experience for their customers.
(12) Example embodiments of the disclosure may provide a recompressor system that configures dynamic Video Ad Serving Template (VAST) tags enabling seamless delivery of mezzanine quality video ads of the highest resolution compatible with the user's system requirements. The system may be configured to receive a VAST tag, parse the VAST tag to identify the location of the video of highest resolution, accessing the video of highest resolution and forwarding it to a recompression engine, which re-compresses the video in one or more formats and sends back location(s) of the videos. The system may receive back one or more video locations and an updated VAST tag and forwarding that for dynamic VAST tag reconfiguration.
(13) With example disclosed embodiments, a video recompression system can be configured provide mezzanine compression, while reducing the video bitrate (bandwidth) for encoded video ads. Aspects of the disclosure can enable encoded video-based ads to be customized for performance. In some embodiments, the video-based ads can be customized for specific devices and/or for specific bandwidth conditions. In some embodiments, the video-based ads can be customized for optimal performance.
Example Processes and Systems
(14) FIG. **1** illustrates an example employment of a high-level architecture **100** for an online video advertisement system. The architecture **100** includes the Brand **101** paying for the advertising. The Agency **102** is who is buying the advertising on behalf of a Brand, Agency operations can be contained within a Brand. The Advertisement Network **103** is used by Agencies to purchase advertising. Advertisers use the Advertisement Server **104** for their advertisements. The demand side platform **105** is used by the buyers. Publishers are able to sell advertising via the Advertisement network for sellers **107**. Publishers will use the advertisement server for sellers **108**. In addition, the architecture **100** includes the publisher **109**, the customer's device **110** and the consumer **111**.
(15) VAST XMLs and their URLs are used to communicate between the Band **101** through the consumer **111**. The methods disclosed are applicable to the Brand **101** through figure the consumer

device **110** as described.

(16) FIG. **2** illustrates an architecture **200** depicting how digital advertising works without the disclosed methods. The consumer **201** may use a device **202**, this may be any display device with computational abilities and protocol support, including televisions, computers, tablets, phone, and watches the consumer loads a website, or an application, and that calls a publisher's advertisement server. The new methods rewriter **203**, the new methods cache for rewriter **204**, the new methods content delivery network (CDN) **205**, and the new methods recompressor **206**, all of which are disclosed herein, are not being used in this scenario. The architecture **200** also includes the supply advertisement server **207**, the demand advertisement server **208**, and the demand CDN **209**.

(17) The architecture **200** depicted in FIG. **2** may include computational processing flow, starting with designation A **210**. At process A **210**, a consumer loading a page may be computationally processed in an application or in a website. At depiction B **211**, the device performs an hypertext transfer protocol (HTTP) GET request **212** for an advertisement from an advertisement supply server. Depiction C **212** indicates a bid on a joined auction. Indication D **213** indicates the win or lose notice of the auction **212**. If the auction is lost, depiction E **214** indicates that the bid may be rebid again. If the auction is won, depiction F **215** will send a video advertisement serving template (VAST) uniform resource locator (URL). At depiction G **216***a-b*, the VAST URL macros are filled in on the server. At depiction H **217**, the VAST URL is returned to the device **202**. Depiction I **218** fills in additional macros on the VAST UL on the device **202**. Next, depiction J **219** returns the VAST XML. Depiction K **220** gets an advertisement stream with an advertisement URL, then depiction L **221** returns the advertisement stream. And lastly, depiction M **222** illustrates that the consumer **201** will view the advertisement stream.

(18) FIG. **3** illustrates an architecture **300** depicting how disclosed methods work in a cache miss scenario. A cache miss occurs when a video advertisement is processed for the first time. The consumer **301** may use a device **302**. The device **302** may be any display device with computational abilities and protocol support, including televisions, computers, tablets, phones, and watches. The consumer **301** loads a website, or an application that calls a publisher's advertisement server. The new methods rewriter **303**, the new methods cache for rewriter **304**, the new methods content delivery network (CDN) **305**, and the new methods recompressor **306**, all of which are disclosed herein, are not being used in this scenario. The architecture **300** also includes a supply advertisement server **307**, a demand advertisement server **308** and a demand CDN **309**.

(19) The architecture **300** depicted in FIG. **3** may include processing flow, starting with designation A **310** which indicates a consumer loading a page in an application or in a website. At depiction B **311**, the device performs an hypertext transfer protocol (HTTP) GET request **312** for an advertisement from an advertisement supply server. Depiction C **312** indicates a bid on a joined auction. Indication D **313** indicates the win or lose notice of the auction **312**. If the auction is lost, depiction E **314** indicates that the auction will stop. If the auction is won, depiction F **315** will send a VAST URL. Depiction G **316** will initiate an application program interface (API) call to the recompressor with null macros. Depiction H **317** puts the customer VAST URL in the cache, and updates it if it is already there. Depiction I **318** indicates a response with EuclidIQ advertisement identification. Depiction J **319** creates a new VAST URL with EuclidIQ Cache universal unique identifier (UUID) and null macros as arguments. Depiction K **320** indicates an API response with a new VAST tag. Depiction L **321***a-b* fills in the macros, both by the supply advertisement server and the client after the following depiction M **322**. Depiction M **322** indicates the EuclidIQ VAST URL returned to the client.

(20) Continuing on FIG. **3**, depiction N **323** indicates the HTTP GET VAST URL. Depiction O **324** reads the cache with Cache UUID. Depiction P **235** looks up the identification to get the original VAST URL. Depiction Q **326** returns the cache data and copies the macros to the original VAST URL. Depiction R **327** copies macros from depiction L **321** to the original VAST URL so that the demand server may be called. Depiction S **328** gets the URL. Depiction T **329** is the response with

the original XML to find the mezzanine URL, while depiction S **328** is the GET URL. Depiction T **329** is the response with the original XML, with the new advertisement video. Depiction U **330** is the process that the VAST XML uses to find the mezzanine URL. Depiction V **331** reads the cache to get the replacement videos with the mezzanine URL.

(21) Continuing on FIG. **3**, depiction AB **332** is the cache data, the cache is empty and therefore indicates a cache miss. Depiction AC **333** is the response to the original VAST XML from depiction T **329**, then there is a pass and uses the original VAST XML, not the new VAST XML in this advertisement request. Depiction AD **334** is the message que to compress asynchronously. Depiction AE **335** gets the advertisement stream, while depiction AF **336** is the advertisement stream added, and starting recompression of the advertisement. Depiction AG **337** posts the advertisement stream to the CDN. Depiction AH **338** updates the cache with the new encode video, the cache key is the mezzanine URL. Depiction AI **339** performs a HTTP GET advertisement stream, which is the original advertisement video because this is a cache miss. Depiction AJ **340** streams the original uncompressed advertisement, the URL to the original stream comes from depiction AC **335**. Next, depiction AK **341** indicates that the consumer sees the original advertisement. The next advertisement request, at depiction A **310** occurs after the advertisement is done with the compression processing. Depiction AF **336** will have a cache hit.

(22) FIG. **4** illustrates how disclosed methods work in a cache hit scenario. The consumer **401** uses the device **402**. This may be any display device with computational abilities and protocol support, including televisions, computers, tablets, phones, and watches. The consumer loads a website, or an application that calls a publisher's advertisement server. The new methods rewriter **403**, the new methods cache for rewriter **404**, the new methods content delivery network (CDN) **405**, and the new methods recompressor **406**, all of which are disclosed herein, are not being used in this scenario. The architecture **300** also includes a supply advertisement server **407**, a demand advertisement server **408** and a demand CDN **409**.

(23) The architecture **400** depicted in FIG. **4** also includes processing flow, starting with designation A **410** which indicates a consumer loading a page in an application or in a website. At depiction B **411**, the device performs an hypertext transfer protocol (HTTP) GET request for an advertisement from an advertisement supply server. Depiction C **412** indicates a bid on a joined auction. Indication D **413** indicates the win or lose notice of the auction **412**. If the auction is lost, depiction E **414** indicates that the auction will stop. If the auction is won, depiction F **415** will send a VAST URL. Depiction G **416** is the API which calls the ReCompressor with null macros. Depiction H **417** puts customer VAST URL in the cache, and updates it if it is already there. The Cache consists of UUID, VAST with Null Macros. Depiction I **418** is the response with EuclidIQ advertisement identification. Depiction J **419** creates a new VAST URL with EuclidIQ Cache UUID and null macros as arguments. Depiction K **420** indicates the API response with a new VAST URL. Depiction L **421** fulls in macros, both by the supply advertisement server and the client after depiction M **422**. Depiction M **442** indicates that the EuclidIQ VAST URL is returned to the client.

(24) Continuing on with FIG. **4**, depiction N **423** is the HTTP GET VAST URL request. Depiction O **424** reads the cache with the cache UUID. Depiction P **425** looks up the identification to get the original VAST URL. Depiction Q **426** indicates the return cache data, and copies macros to the original VAST URL. Depiction R **427** copies macros from depiction L **421** to the original VAST URL so that the demand server may be called. Depiction S **428** gets the VAST URL. Depiction T **429** is the response with the original XML with the new advertisement video. Depiction U **430** is the process that the VAST XML uses to find the mezzanine URL. Depiction V **431** reads the cache to get the replacement video with the mezzanine URL. Depiction W **432** returns the cache data. Depiction X **433** rewrites the VAST XML with the replacement video. Depiction Y **434** is the response with the VAST XML. Depiction Z **435** is the HTTP GET request for the advertisement stream. Depiction AL **436** is the streaming advertisement. Next, depiction AM **437** indicates that

the consumer sees the recompressed, reformatted, repackaged advertisement that was cached.
(25) Digital Processing Environment
(26) Example implementations of the present invention may be implemented in a software, firmware, or hardware environment. FIG. **5** illustrates one such environment. For example, the method/system described above (including FIGS. **1-4**) may be implemented in the environment of FIG. **6**. Client computer(s)/devices **501** (e.g., mobile phones or computing devices) and a cloud **502** (or server computer or cluster thereof) provide reencoding, VAST tag intercepting, VAST tag reconfiguring, processing, storage, encoding, decoding, and input/output devices executing application programs and the like.
(27) Client computer(s)/devices **501** can also be linked through communications network **70** such as the internet and a content delivery network (CDN) to other computing devices, including other client devices/processes **501** and server computer(s) **502**. This communications network framework may include ad networks, ad servers, and demand side platforms, ad server for buyers, or the demand side platform, or the supply side platform, or the ad server for sellers. Communications network **500** can be part of a remote access network, a global network (e.g., the Internet), a worldwide collection of computers, Local area or Wide area networks, and gateways that currently use respective protocols (TCP/IP, Bluetooth, etc.) to communicate with one another. Other electronic devices/computer network architectures are suitable.
(28) Embodiments of the invention may include means for reencoding, recompressing, tracking, tracking, VAST tag configuring, modeling, filtering, tuning, decoding, streaming, or displaying video or data signal information. FIG. **6** is a diagram of the internal structure of a computer/computing node (e.g., client processor/device/mobile phone device/tablet **501** or server computers **502**) in the processing environment of FIG. **5**, which may be used to facilitate encoding such videos or data signal information. For example, method/system described here (including FIGS. **1-4**) may be implemented by the computer/computing node of FIG. **6**.
(29) Each computer **501**, **502** contains a system bus **601**, where a bus is a set of actual or virtual hardware lines used for data transfer among the components of a computer or processing system. Bus **601** is essentially a shared conduit that connects different elements of a computer system (e.g., recompressor, processor, encoder chip, decoder chip, disk storage, memory, input/output ports, etc.) that enables the transfer of data between the elements. Attached to the system bus **601** is an I/O device interface **602** for connecting various input and output devices (e.g., keyboard, mouse, displays, printers, speakers, etc.) to the computer **501**, **502**. Network interface **603** allows the computer to connect to various other devices attached to a network (for example, the network illustrated at **500** of FIG. **5**). Memory **604** provides volatile storage for computer software instructions **604** and data **605** used to implement a software implementation of the present invention (e.g., codec: recompressor/encoder/decoder).
(30) Disk storage **606** provides non-volatile storage for computer software instructions **605** (equivalently “OS program”) and data **606** used to implement an embodiment of the present invention: it can also be used to store the video in compressed format for long-term storage. Central processor unit **608** is also attached to system bus **601** and provides for the execution of computer instructions. Note that throughout the present text, “computer software instructions” and “OS program” are equivalent.
(31) In one example, a recompressor or other encoding system may be configured with computer readable instructions **606** that implement a method and/or system designed to predict the target (encoding) to deploy to the user device that will provide the fastest rate of transmission to the user device (the consumer/consuming device). In an optional embodiment, an encoder or other recompressor/encoder/decoder system may be configured with computer readable instructions **605** that implement a method and/or designed to select videos from nodes in the content delivery network (CDN) that are in close proximity to speed up transmission to the user device. In another optional example, a recompressor, an encoder or other encoding system may be configured with

computer readable instructions **605** that implement a method and/or system designed to determine an optimal subset of perceptual quality metrics for constrained bitrate encoding applications for the user device. In another optional example, a recompressor system may be configured with computer readable instructions **605** that implement a method and/or system designed to determine an optimal number of B-frames for encoding a video.
(32) In one embodiment, the processor routines **605** and data **606** are a computer program product, with an recompressor (generally referenced **605**), including a computer readable medium capable of being stored on a storage device **606** which provides at least a portion of the software instructions for the recompressor/encoder/decoder.
(33) The computer program product **605** can be installed by any suitable software installation procedure, as is well known in the art. In another embodiment, at least a portion of the encoder software instructions may also be downloaded over a cable, communication, and/or wireless connection. In other embodiments, the recompressor system software is a computer program propagated signal product embodied on a nontransitory computer readable medium, which when executed can be implemented as a propagated signal on a propagation medium (e.g., a radio wave, an infrared wave, a laser wave, a sound wave, or an electrical wave propagated over a global network such as the Internet, or other network(s)). Such carrier media or signals provide at least a portion of the software instructions for the present invention routines/program **605**.
(34) In alternate embodiments, the propagated signal is an analog carrier wave or digital signal carried on the propagated medium. For example, the propagated signal may be a digitized signal propagated over a global network (e.g., the Internet), a telecommunications network, or other network. In one embodiment, the propagated signal is transmitted over the propagation medium over a period of time, such as the instructions for a software application sent in packets over a network over a period of milliseconds, seconds, minutes, or longer. In another embodiment, the computer readable medium of computer program product **605** is a propagation medium that the computer system **501** may receive and read, such as by receiving the propagation medium and identifying a propagated signal embodied in the propagation medium, as described above for the computer program propagated signal product.
(35) In an optional embodiment, example recompression/encoding techniques employed by the recompressor may be provided by the encoding techniques disclosed in the following which are incorporated by reference in their entirety: U.S. application Ser. No. 17/144,924, filed Jan. 8, 2021, now U.S. Pat. No. 11,350,105, issued May 31, 2022, which is a continuation of U.S. application Ser. No. 16/926,089, filed Jul. 10, 2020, now U.S. Pat. No. 11,159,801 issued Oct. 26, 2021, which is a continuation of U.S. application Ser. No. 16/420,796, filed May 23, 2019, now U.S. Pat. No. 10,757,419, issued Aug. 25, 2020, which is a continuation-in-part of International Application No. PCT/US2017/067413, which designated the United States and was filed on Dec. 19, 2017, and which claims the benefit of U.S. Provisional Application No. 62/452,265 filed on Jan. 30, 2017.
(36) The teachings of all patents, published applications and references cited herein are incorporated by reference in their entirety.
(37) While example embodiments have been particularly shown and described, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the embodiments encompassed by the appended claims.

Claims

1. A computer-implemented method of compressing and video media to a user device, the method comprising: loading, by a consumer, a video media on a device that performs a HTTP GET for an advertisement from an advertisement server; joining, by a demand side server an advertisement auction; bidding on and winning, by a demand side server, the advertisement auction; transmitting, via the demand side server or a supply side advertisement server, or any server transmitting video

media to client devices, a recompression server API; passing a VAST URL to a recompression server and adding or updating a record in a cache with a key value pair; building and storing, by the recompression server, a new VAST XML and responding with a new VAST URL to the new VAST XML; transmitting, via the demand side server or supply side advertisement server, or any server transmitting video media, and filling in URL macro fields; transmitting, via the demand side server or supply side advertisement server, or any server transmitting video media, and responding to the device with a VAST URL; filling in, by the device, the macros on VAST URL and the HTTP headers with hardware and software information during a HTTP GET; performing, by the device, a HTTP GET on the VAST URL with macros; performing, via the VAST rewriter server, a check of the cache for a compressed ad with a key value pair; copying, by the server, macros to original VAST URL, enabling the demand server to be called; responding, via the demand advertisement server, with the original XML and/or a new advertisement video; processing, via the VAST rewriting server, a VAST XML, and finding the mezzanine URL for recompression; reading, via the VAST rewriting server, the cache to receive the replacement videos with the mezzanine URL; determining, via the VAST rewriting server, if the cache is empty; responding, via the VAST rewriting server, to the device with the original VAST XML and using the original VAST XML in the advertisement request; performing, via the VAST rewriting server, a message queue to compress asynchronously; getting, via the recompression server HTTP, the advertisement stream to be recompressed; responding, via the demand CDN, with the advertisement stream and the recompression server starting compression of the advertisement; posting, via the recompression server, the newly compressed advertisement stream(s) to the CDN; updating, via the recompression server, the cache with the newly encoded video data with a key value pair; performing, via the device, a HTTP GET advertisement stream with original advertisement video; and streaming, via the device, the original uncompressed advertisement, enabling the consumer to view the original advertisement.

2. The method of claim 1, wherein the VAST XML regenerating process comprises of original VAST XML data, new VAST XML data, and original video while the new video is recompressed, reformatted, repackaged or transformed from on codec to another to be used in future ad requests.

3. The method of claim 1, wherein the VAST XML regenerating process comprises of original VAST XML data, new VAST XML data, and reformatted video, with different frame sizes, different bit rates, different codec, different containers, different stream protocols.

4. The method of claim 1, further comprising, VAST XML or VPAID Script or like protocol is used to communicate between components.

5. The method of claim 1, further comprising, various video and audio compression, packaging, and streaming protocols.

6. The method of claim 1, further comprising processing multiple unique ad videos associated with a single VAST/VPAID object.

7. A computer-implemented method of compressing and delivering video an ad or general video to a user device, comprising: a consumer loading media on a device that performs a HTTP GET for an ad from an ad server; demand side server, optionally, joins an ad auction and bids and wins the auction; demand side server or supply side ad server or any server transmitting any kind of video to client devices calls a recompression server API and passes a VAST URL to a recompression server and that server adds or updates a record in a cache with a key value pair; recompression server builds and stores a new VAST XML and responds with a new VAST URL to the new VAST XML; demand side server or supply side ad server or any server transmitting any kind of video fills in URL macro fields; demand side server or supply side ad server or any server transmitting any kind of video responds to the device with a VAST URL; device fills in macros on VAST URL; device fills in the HTTP headers with hardware and software information during a HTTP GET; device performs a HTTP get on the VAST URL with macros; VAST rewriter server performs a check of the cache for a compressed ad with a key value pair; server copies macros to original VAST URL

so that the demand server may be called; demand ad server responds with original XML and optionally a new ad video; and VAST rewriting server processes a VAST XML to find the mezzanine URL for recompression; VAST rewriting server reads the cache to get the replacement videos with the mezzanine URL; VAST rewriter server receives the cache data; VAST rewriter uses the HTTP header info about the device's hardware and software capabilities to determine the optimum audio and video set respond with and server generates a new VAST XML with the replacement video; VAST rewriter adds new XML, URLs and Scripting codes to the new vast tag for the purpose of video start, video impression, video clicks tracking and bandwidth determination; this information is used to optimize audio and video settings in subsequent advertisements for this device, spatially and temporally because connection speed varies spatially and temporally; VAST rewriter dynamically changes what is included in the new XML, URLs and Scripting codes for the purpose of measuring video, network and device performance and applying statistical quantitative techniques; VAST rewriter server response with VAST XML to the client; and Client HTTP GETs the ad stream; Client streams ad; and Consumer sees ad, the recompressed, reformatted, repackaged ad that was cached.

8. The method of claim 7, wherein the VAST XML regenerating process comprises of original VAST XML data, new VAST XML data, and ad video new video recompressed, reformatted, repackaged or transformed from on codec to another.

9. The method of claim 7, wherein the HTTP header information about the device hardware and software is used to dynamically prepare the optimum audio video in terms of frame sizes, bit rates, codec, containers, streaming protocols.

10. The method of claim 7, wherein the VAST XML regenerating process comprises of original VAST XML data, new VAST XML data, and reformatted video, with different frame sizes, different bit rates, different codec, different containers, different stream protocols.

11. The method of the claim 7, wherein the VAST XML and VPAID Script or like protocols are used to dynamically add additional video start, video impression, video click tracking and how this information can be used to optimize subsequent ads to further optimized.

12. The method of claim 7, further comprising, VAST XML or VPAID Script or like protocol is used to communicate between components.

13. The method of claim 7, further comprising, various video and audio compression, packaging, and streaming protocols.

14. The method of claim 7, further comprising processing multiple unique ad videos associated with a single VAST/VPAID object.

15. A codec system comprising; a client device configured to communicate via a session handler with at least one server in a content delivery network; the at least one server configured to deploy a first video encoding to the client device via a first VAST tag transmitted via the session handler to the client device; a recompressor system configured to process the first VAST tag to identify the first video encoding; the recompressor system configured to recompress the first video encoding using bitrate encoding quality metrics to dynamically optimize the first video encoding for the client device such that the first video encoding is transformed into a first mezzanine encoding; the recompressor system configured to dynamically reconfigure the first VAST tag resulting in a transformed vast tag identifying the first mezzanine encoding while preserving information about the client device from the VAST tag including existing VAST and VPAID data elements so that all tracking information is preserved; and the at least one server configured to respond to a request to deploy the first video encoding to the client by verifying that the first video encoding is in the video cache of the CDN and responding by forwarding the transformed VAST tag identifying the first mezzanine encoding to the client device via the session handler; and the client device be configured to respond to the transformed VAST tag by processing at least a portion of the first mezzanine encoding via the transformed VAST tag.

16. The codec system of claim 15 wherein recompressor system identifies ad proxy server in the

CDN having a geographic location near a physical location of the client device to improve content delivery speed at the client device of the first video encoding with a highest resolution resulting in the first mezzanine encoding.

17. The codec system of claim 16 wherein the recompressor system is stored on a non-transitory computer readable medium, the recompressor system is configured to be executed by at least one computer processor, causing the computer processor to perform bitrate encoding of the first video encoding, where the bitrate encoding includes determining an optimal quality metrics for bitrate encoding, the recompressor optionally performing bitrate encoding on the first video encoding by: computationally processing a plurality of perceptual quality metrics, a plurality of statistical models, a desired mean opinion score (MOS), and a mean opinion score (MOS) prediction error limit; deriving, from perceptual quality metrics, a first subset of the plurality of statistical models, and first subset of at least one the initial encodings, a first derived model that predicts mean opinion scores (MOS); using the first derived model, the first subset of the plurality of perceptual quality metrics, the first subset of the plurality of statistical models, and a first subset of the initial encodings, computationally determining first model bitrates that are proximate to the desired mean opinion score (MOS); computing, for the first subset of the plurality of the perceptual quality metrics, for the first subset of the plurality of statistical models, and for the first video encoding, a first subset of bitrate errors representing a difference between a respective model bitrate and the respective corresponding true bitrate of the first video encoding; computing, for the first subset of the plurality of perceptual quality metrics and the first subset of the plurality of statistical models, an average bitrate savings across at least a portion of the encoded true bitrate videos, the average bitrate savings calculated by comparing the respective model bitrate to the respective corresponding true bitrate of first video encoding; comparing respective further subsets from the plurality of perceptual quality metrics and the plurality of statistical models with the first subset of the plurality of perceptual quality metrics and the first subset of the plurality of statistical models, respectively, to determine an optimal subset of at least one of the perceptual quality metrics from the plurality of the perceptual quality metrics and at least one of the plurality of statistical models, which produce the largest average bitrate savings, while incurring substantially no absolute mean opinion scores (MOS) prediction errors greater than the mean opinion scores (MOS) prediction error limit; and wherein, the optimal subset produces a predicted mean opinion scores (MOS) close to the desired mean opinion scores (MOS) and maximizes the average bitrate savings subject to a constraint on the mean opinion scores (MOS) prediction error being less than the mean opinion score (MOS) prediction error limit.

18. The codec system of claim 16, wherein the plurality of statistical models are derived from at least one of: machine learning or statistical modeling.